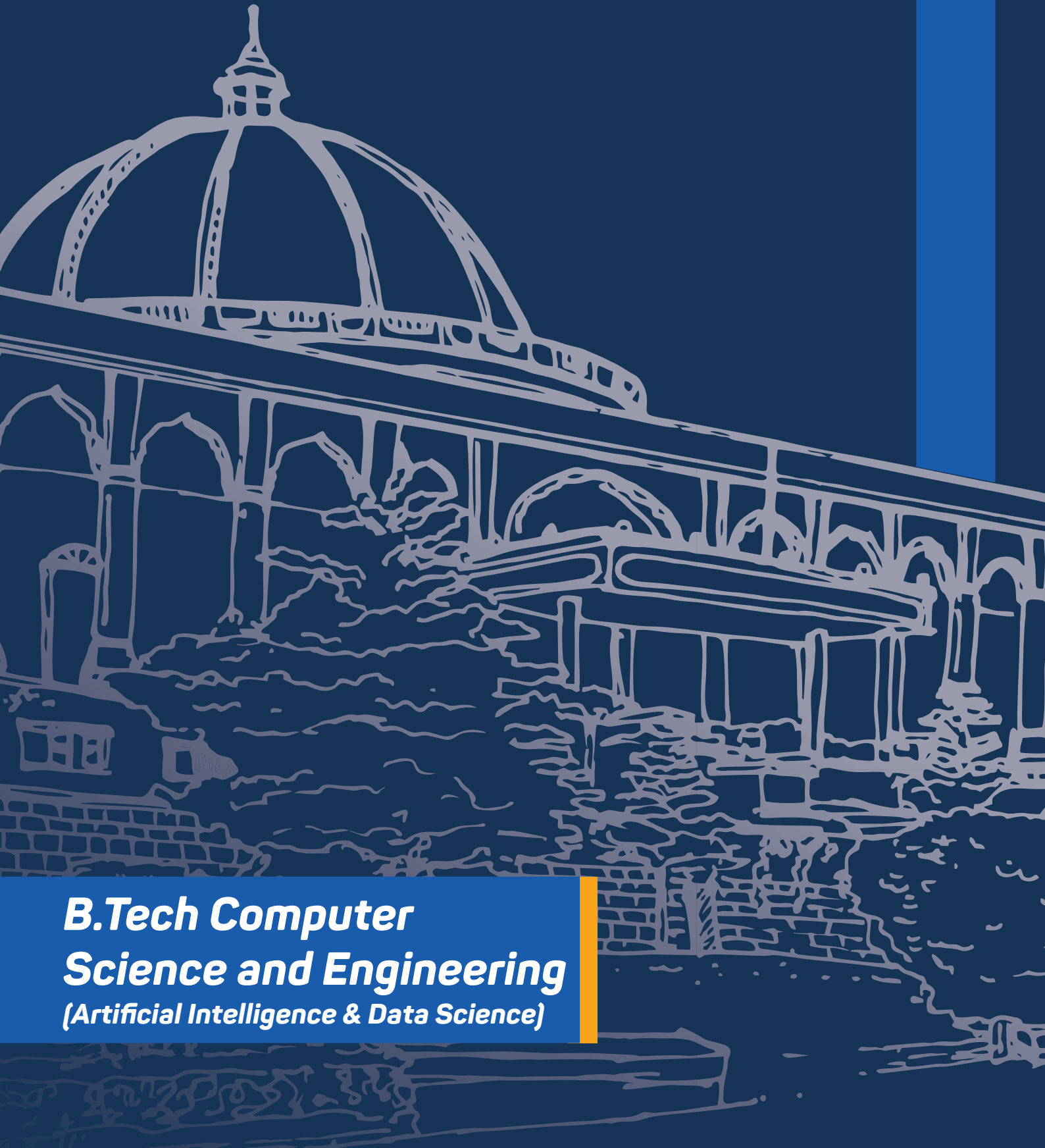




***B.Tech Computer
Science and Engineering***
(Artificial Intelligence & Data Science)

Division	Faculty of Engineering and Technology
School Name	School of Computer Science & Engineering
Department Name	Department of Computer Engineering and Technology
Programme Name	B.Tech Computer Science and Engineering (Artificial Intelligence & Data Science)

Semester	Odd (I)	Even (II)	Total Credits
First Year	20	21	41
Second Year	22	22	44
Third Year	22	22	44
Fourth Year	22	14	36

***** COURSE BASKET *****

Course Type	Description
Programme Core	Courses dealing with foundations, depth and breadth of the major in which student is admitted at MIT-WPU
Programme Electives	Open electives under the programme allow students to specialise in a particular area connected to their major.
University Core	Courses that reflect the core MIT-WPU values and the mission of Life Transformation of students.
University Electives	Multidisciplinary courses across the faculties at MIT-WPU and outside the programme core.

Semester I

Semester	Course Type	Course Name / Course Title	Total Credits
I	UC	Indian Constitution	1
I	UC	Environment and Sustainability	1
I	UC	Yoga - I	1
I	UC	Social Leadership Development Program	1
I	UC	Financial Literacy	1
I	UC	Python Programming	2
I	PF	Linear Algebra and Differential Calculus	3
I	PF	Engineering Chemistry	3
I	PF	Engineering Physics	3
I	PF	Engineering Graphics	3
I	PF	Ideas and Innovations in Manufacturing	1
		Total Credits:	20

Semester II

Semester	Course Type	Course Name / Course Title	Total Credits
II	UC	Yoga - II	1
II	UC	Co-creation	1
II	UC	Foundation of Peace	2
II	UC	Indian Knowledge System (General)	2
II	UC	Sports	1
II	PF	Engineering Mechanics	3
II	PF	Foundations of Programming	4
II	PF	Discrete Mathematics with Graph Theory	3
II	PF	Foundations of Computer Architecture and System Design	4
		Total Credits:	21

Semester - III

Semester	Course Type	Course Name / Course Title	Total Credits
III	UC	Spiritual and Cultural heritage: Indian Experience	2
III	UC	Research Innovation Design Entrepreneurship (RIDE)	1
III	PF	Calculus and Numerical Methods	4
III	PF	Data Structures	3
III	Program Capstone Project/Seminar and Internships	Data Structures Laboratory	1
III	Program Capstone Project/Seminar and Internships	Web Technology Laboratory	1
III	PM	Database Management Systems	3
III	PM	Database Management Systems Laboratory	1
III	PM	Software Engineering and Modelling	2
III	PM	Software Engineering and Modelling Lab	1
III	UE	University Electives	1
		Total	20

SEMESTER - IV

Semester	Course Type	Course Name / Course Title	Total Credits
IV	UC	Rural Immersion	1
IV	UC	Life Transformation Skills	1
IV	UE	University Electives - II	3
IV	PF	Probability and Statistics	4
IV	PM	Data Engineering Concepts	2
IV	PM	Data Engineering Concepts Laboratory	1
IV	PF	Artificial Intelligence and Expert Systems	2
IV	Program Capstone Project/ Seminar and Internships	Artificial Intelligence and Expert Systems Laboratory	1
IV	PM	Advanced Data Structures	3
IV	Program Capstone Project/ Seminar and Internships	Advanced Data Structures Laboratory	1
IV	PM	Object Oriented Programing Laboratory	1
IV	Program Capstone Project/ Seminar and Internships	Cognitive Skills	1
IV	Program Capstone Project/ Seminar and Internships	Seminar	1
		Total Credits:	22

Semester - V

Semester	Course Type	Course Name / Course Title	Total Credits
V	UC	Managing Conflicts Peacefully: Tools and Techniques	2
V	UE	University Electives - III	3
V	PM	Data Visualization	1
V	PM	Machine Learning	3
V	PM	Machine Learning Lab	1
V	PM	Operating Systems	3
V	PM	Operating Systems Laboratory	1
V	PM	Theory of Computation	3
	PF	Embedded Systems and IOT Laboratory	1
	PE	Program Elective - I	4
		Total Credits:	22

Semester - VI

Semester	Course Type	Course Name / Course Title	Total Credits
VI	UC	National Academic Immersion Program	2
VI	PE	Program Elective - II	4
VI	PM	Deep Learning	3
VI	PM	Deep Learning Laboratory	1
VI	PM	Cognitive Computing and Natural Language Processing	3
VI	Program Capstone Project/ Seminar and Internships	Cognitive Computing and Natural Language Processing Lab	3
VI	Program Capstone Project/ Seminar and Internships	Tools for Data Science	1
VI	Program Capstone Project/ Seminar and Internships	Competitive Coding	1
VI	Program Capstone Project/ Seminar and Internships	Mini Project using Java Programming	1
VI	PM	Design and Analysis of Algorithm	3
		Total Credits:	22

Semester - VII

Semester	Course Type	Course Name / Course Title	Total Credits
VII	Program Major	Bigdata Technologies	4
VII	Program Major	Distributed and Cloud Computing	4
VII	Program Capstone Project/Seminar and Internships	Capstone Project	6
VII	Program Major	Network and Cyber Security	4
VII	Program Electives	Program Elective - III	4
		Total Credits:	22

Semester - VIII

Semester	Course Type	Course Name / Course Title	Total Credits
VIII	PE	Program Elective - IV	4
VIII	Program Capstone Project/Seminar and Internships	Internship	10
		Total Credits:	14

Semester	Course Type	Course Name / Course Title	Total Credits
V	PE - I	A. User Interface and User Experience Design	4
V	PE - I	B. Graph Machine Learning	4
V	PE - I	C. Geospatial Data Analysis	4
V	PE - I	D. AI Systems and Applications	4
VI	PE - II	A. Computer Vision	4
VI	PE - II	B. Augmented Reality and Virtual Reality	4
VI	PE - II	C. Edge Artificial Intelligence	4
VI	PE - II	D. Soft Computing	4
VII	PE - III	A. Generative Adversarial Network	4
VII	PE - III	B. Time Series Analysis and Forecasting	4
VII	PE - III	C. Bioinformatics Algorithms and Applications	4
VII	PE - III	D. Blockchain Technology	4
VIII	PE - IV	A. Generative AI	4
VIII	PE - IV	B. Sentiment and Emotion Analysis	4
VIII	PE - IV	C. Quantum Computing	4
VIII	PE - IV	D. Scalable Data Science	4

*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.